Case Study 5 — Hierarchical Bayesian Logistic Regression: Gradient-Based Optimisation and Posterior Variance Collapse

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Abstract

This report applies the gradient-based optimisation methods from ENEL445 to variational inference in hierarchical Bayesian logistic regression. This model combines two sources of analytical difficulty from previous case studies: the non-conjugate logistic likelihood (Case Study 3) and the hierarchical random-effects structure (Case Study 4). Group random effects $u_j \sim \mathcal{N}(0, \tau_u^{-1})$ augment the fixed effects $\boldsymbol{\beta}$, and the Jaakkola–Jordan (Jaakkola & Jordan, 2000) bound is required to retain a tractable ELBO. The unconstrained parameter dimension is $D = 17$.

Four optimisation methods are applied: CAVI, gradient ascent with Armijo backtracking, Newton's method, and BFGS. The reference posterior is provided by a Pólya–Gamma blocked Gibbs sampler (Polson et al., 2013) with $T = 50$ truncation terms, 3 chains $\times$ 2000 iterations.

A key finding is pronounced posterior variance collapse in both β_0 and β_1 (SD ratios ≈ 0.46), reflecting the mean-field VI family's failure to capture the posterior correlation between the fixed effects and the group random effects. CAVI and BFGS converge to the same ELBO optimum 1050$\times$ and 81$\times$ faster than the PG Gibbs sampler respectively. Newton's method again fails to find the correct optimum, converging to a suboptimal point with markedly worse ELBO.

Introduction

Hierarchical logistic regression is a natural model for binary outcomes observed in grouped or clustered data: students within schools, patients within hospitals, or binary responses from repeated measurements on the same subject. Each group j has a random intercept u_j that shifts the group-specific log-odds relative to the population, while $\boldsymbol{\beta}$ captures the population-level covariate effect.

Bayesian inference in this model requires integrating over both $\boldsymbol{\beta}$ and the random effects $\mathbf{u} = (u_1, \ldots, u_J)^T$, and over the random-effect precision τ_u. The logistic likelihood is not conjugate to any natural prior, so both MCMC and variational approximations are computationally more demanding than the Gaussian case.

This report is Case Study 5 in this project, combining the Jaakkola–Jordan bound from Case Study 3 with the hierarchical structure from Case Study 4. The mean-field ELBO is maximised using four optimisation strategies, and the results are compared against the Pólya–Gamma Gibbs reference.

1 Model Formulation

1.1 Hierarchical Bayesian Logistic Regression

The model is

$$
\begin{aligned}
y_{ij} \mid \boldsymbol{\beta}, u_j &\sim \text{Bernoulli}\big(\sigma(\mathbf{x}_{ij}^T\boldsymbol{\beta} + u_j)\big), \quad i = 1, \ldots, n_j,\ j = 1, \ldots, J, && (1)\\
u_j \mid \tau_u &\sim \mathcal{N}(0,\ \tau_u^{-1}), && (2)\\
\boldsymbol{\beta} &\sim \mathcal{N}(\mathbf{0},\ \lambda^{-1}\boldsymbol{I}_p), \quad \lambda = 0.1, && (3)\\
\tau_u &\sim \text{Gamma}(\alpha_u,\ \gamma_u), \quad \alpha_u = 1,\ \gamma_u = 0.1, && (4)
\end{aligned}
$$

where $\sigma(\psi) = (1 + e^{-\psi})^{-1}$ is the logistic sigmoid and $\mathbf{x}_{ij} = (1, x_{ij})^T \in \mathbb{R}^2$.

The test case uses $N = 150$ binary observations in $J = 5$ groups (30 per group), $x_{ij} \sim \text{Uniform}(-2, 2)$, and true parameters $\boldsymbol{\beta}_{\text{true}} = (-1.0,\ 2.0)$, $\tau_{u,\text{true}} = 2.0$ ($\sigma_u \approx 0.707$).

1.2 Mean-Field Variational Approximation

The mean-field variational family is

$$
q(\boldsymbol{\beta}, \mathbf{u}, \tau_u) = q(\boldsymbol{\beta}) \prod_{j=1}^{J} q(u_j) \ \cdot\ q(\tau_u), \tag{5}
$$

with

$$
\begin{aligned}
q(\boldsymbol{\beta}) &= \mathcal{N}(\boldsymbol{\mu}_\beta,\ \boldsymbol{\Sigma}_\beta), \quad \boldsymbol{\Sigma}_\beta = \boldsymbol{L}\boldsymbol{L}^T, && (6)\\
q(u_j) &= \mathcal{N}(m_j,\ v_j), && (7)\\
q(\tau_u) &= \text{Gamma}(a_u,\ b_u). && (8)
\end{aligned}
$$

1.3 Jaakkola–Jordan ELBO

The log-likelihood $\mathbb{E}_q[\log \sigma(\psi_{ij})]$ where $\psi_{ij} = \mathbf{x}_{ij}^T\boldsymbol{\beta} + u_j$ is intractable under a Gaussian q. The Jaakkola–Jordan bound introduces local variational parameters $\xi_{ij} > 0$:

$$
\log \sigma(\psi_{ij}) \geq \tfrac{\psi_{ij}}{2} - \log 2\cosh\tfrac{\xi_{ij}}{2} - \tfrac{\xi_{ij}}{2}\left(\mathbb{E}_q[\psi_{ij}^2]/\xi_{ij}^2 - 1\right)\tfrac{\tanh(\xi_{ij}/2)}{4}. \tag{9}
$$

At the optimal $\xi_{ij}^* = \sqrt{\mathbb{E}_q[\psi_{ij}^2]}$, where

$$\mathbb{E}_q[\psi_{ij}^2] = (\mathbf{x}_{ij}^T \boldsymbol{\mu}_\beta + m_j)^2 + \mathbf{x}_{ij}^T \boldsymbol{\Sigma}_\beta \mathbf{x}_{ij} + v_j, \tag{10}$$

the ELBO is

$$\begin{aligned}\mathcal{L}(\boldsymbol{\theta}) = \sum_{ij} \big[(y_{ij} - \tfrac{1}{2})\bar{\psi}_{ij} - \log 2 - \log\cosh \tfrac{\xi_{ij}^*}{2}\big] - \mathrm{KL}(q(\boldsymbol{\beta}) \| p(\boldsymbol{\beta})) \\ + \mathbb{E}_q[\log p(\mathbf{u} \mid \tau_u)] + \mathbb{E}_q[\log p(\tau_u)] + \sum_j H[q(u_j)] + H[q(\tau_u)],\end{aligned} \tag{11}$$

where $\bar{\psi}_{ij} = \mathbf{x}_{ij}^T \boldsymbol{\mu}_\beta + m_j$ and all terms are available in closed form.

1.4 CAVI Update Equations

Let $\lambda_{ij} = \tanh(\xi_{ij}^*/2)/(4\xi_{ij}^*)$. The coordinate ascent updates are:

$$\boldsymbol{\Sigma}_\beta^{-1} \leftarrow \lambda \boldsymbol{I} + 2 \sum_{ij} \lambda_{ij}\, \mathbf{x}_{ij} \mathbf{x}_{ij}^T, \tag{12}$$

$$\boldsymbol{\mu}_\beta \leftarrow \boldsymbol{\Sigma}_\beta \sum_{ij} (y_{ij} - \tfrac{1}{2} - 2\lambda_{ij} m_j)\, \mathbf{x}_{ij}, \tag{13}$$

$$v_j^{-1} \leftarrow 2 \sum_{i \in j} \lambda_{ij} + \mathbb{E}[\tau_u], \quad m_j \leftarrow v_j \sum_{i \in j} (y_{ij} - \tfrac{1}{2} - 2\lambda_{ij} \mathbf{x}_{ij}^T \boldsymbol{\mu}_\beta), \tag{14}$$

$$a_u \leftarrow \alpha_u + \tfrac{J}{2}, \quad b_u \leftarrow \gamma_u + \tfrac{1}{2} \sum_j (m_j^2 + v_j). \tag{15}$$

1.5 Unconstrained Parameterisation

The unconstrained vector $\boldsymbol{\theta} \in \mathbb{R}^{17}$ is

$$\boldsymbol{\theta} = \big(\boldsymbol{\mu}_\beta,\ \tilde{\ell}_{11}, \ell_{21}, \tilde{\ell}_{22},\ m_1, \ldots, m_5,\ \log v_1, \ldots, \log v_5,\ \log a_u, \log b_u\big), \tag{16}$$

giving $D = 2 + 3 + 5 + 5 + 2 = 17$.

1.6 Key Notation

Symbol	Meaning
Bayesian model	
$\boldsymbol{X}$	Design matrix ($N \times p$), rows $\mathbf{x}_{ij}^T$
$\mathbf{y}$	Binary response vector ($y_{ij} \in \{0, 1\}$)
$\boldsymbol{\beta}$	Fixed-effect coefficient vector
u_j	Group random effect for group j
τ_u	Random-effect precision
$\sigma(\cdot)$	Logistic sigmoid $\sigma(\psi) = (1 + e^{-\psi})^{-1}$
N, J, p	Total observations, groups, predictors
Variational approximation	
$q(\boldsymbol{\beta})$	Variational Gaussian for fixed effects
$\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta$	Mean and covariance of $q(\boldsymbol{\beta})$
$\boldsymbol{L}$	Cholesky factor, $\boldsymbol{\Sigma}_\beta = \boldsymbol{L}\boldsymbol{L}^T$
m_j, v_j	Mean and variance of $q(u_j)$
a_u, b_u	Gamma parameters of $q(\tau_u)$
ξ_{ij}	Local JJ variational parameter; optimal at $\sqrt{\mathbb{E}_q[\psi_{ij}^2]}$
λ_{ij}	JJ weight $\tanh(\xi_{ij}/2)/(4\xi_{ij})$
$\mathcal{L}(\boldsymbol{\theta})$	Jaakkola–Jordan ELBO (maximised)
Optimisation	
$\boldsymbol{\theta}$	Unconstrained vector ($D = 17$)
α_t	Step size at iteration t
Pólya–Gamma sampler	
ω_{ij}	PG latent variable: $\omega_{ij} \mid \boldsymbol{\beta}, u_j \sim \mathrm{PG}(1, \psi_{ij})$
κ_{ij}	Pseudo-response $\kappa_{ij} = y_{ij} - 1/2$
T	Truncation order for PG series ($T = 50$)
Lecture correspondence (Prof. R. S. Smith, ENEL445 2026)	
$J(\theta)$	Cost function $= \mathcal{L}(\boldsymbol{\theta})$ in this report
Φ	Regressor matrix $= \boldsymbol{X}$ in this report

2 Pólya–Gamma Blocked Gibbs Sampler

The Pólya–Gamma (PG) identity of Polson et al. (2013),

$$\sigma(\psi)^y(1 - \sigma(\psi))^{1-y} \propto e^{\kappa\psi} \int_0^\infty e^{-\omega\psi^2/2}\, p(\omega)\, \mathrm{d}\omega, \quad \omega \sim \mathrm{PG}(1, 0), \tag{17}$$

introduces a latent variable ω_{ij} that renders the logistic likelihood conditionally Gaussian. Each Gibbs sweep samples:

$$\omega_{ij} \mid \boldsymbol{\beta}, u_j \sim \mathrm{PG}(1,\ \mathbf{x}_{ij}^T\boldsymbol{\beta} + u_j), \tag{18}$$

$$\boldsymbol{\beta} \mid \boldsymbol{\omega}, \mathbf{u}, \mathbf{y} \sim \mathcal{N}(V_\beta(\boldsymbol{X}^T\boldsymbol{\kappa} - \boldsymbol{X}^T\mathrm{diag}(\boldsymbol{\omega})\mathbf{u}[\mathrm{g}]),\ V_\beta), \quad V_\beta = (\lambda\boldsymbol{I} + \boldsymbol{X}^T\mathrm{diag}(\boldsymbol{\omega})\boldsymbol{X})^{-1}, \tag{19}$$

$$u_j \mid \boldsymbol{\beta}, \boldsymbol{\omega}_j, \tau_u, \mathbf{y} \sim \mathcal{N}(V_j^* m_j^*,\ V_j^*), \quad V_j^* = \left(\textstyle\sum_{i\in j}\omega_{ij} + \tau_u\right)^{-1}, \quad m_j^* = \textstyle\sum_{i\in j}(\kappa_{ij} - \omega_{ij}\mathbf{x}_{ij}^T\boldsymbol{\beta}), \tag{20}$$

$$\tau_u \mid \mathbf{u} \sim \mathrm{Gamma}\!\left(\alpha_u + \tfrac{J}{2},\ \gamma_u + \tfrac{1}{2}\textstyle\sum_j u_j^2\right). \tag{21}$$

PG draws use the truncated series with $T = 50$ terms. Three chains run for 2000 iterations each with 500 burn-in, yielding $3 \times 1500 = 4500$ post-burnin samples.

3 Optimisation Problem Formulation

Decision Variables and Constraints

Optimisation is over $\boldsymbol{\theta} \in \mathbb{R}^{17}$. All constraints ($\boldsymbol{\Sigma}_\beta \succ 0$, $v_j > 0$, $a_u, b_u > 0$) are satisfied automatically by the parameterisation.

Objective Function

Maximise the Jaakkola–Jordan ELBO $\mathcal{L}(\boldsymbol{\theta}) : \mathbb{R}^{17} \to \mathbb{R}$ at optimal $\xi_{ij}^* = \sqrt{\bar{\psi}_{ij}^2 + \mathbf{x}_{ij}^T \boldsymbol{\Sigma}_\beta \mathbf{x}_{ij} + v_j}$.

Optimisation Components

CAVI. Closed-form coordinate updates (Equations 12–above). Unit step; exit when $|\Delta\mathcal{L}| < 10^{-8}$.

Gradient ascent. Direction: central FD gradient ($h = 10^{-5}$). Armijo backtracking, $\alpha_0 = 0.5$, $c_1 = 0.1$, $\rho = 0.5$. Exit: $|\Delta\mathcal{L}| < 10^{-7}$ or 2000 iterations.

Newton's method. FD Hessian (17×17); regularised to ensure ascent direction. Armijo backtracking; exit at 50 iterations.

BFGS. Inverse-Hessian approximation via rank-2 BFGS update. Wolfe conditions (scipy); exit at $\|\boldsymbol{g}\| < 10^{-7}$ or 2000 iterations.

4 Numerical Study

4.1 Test Case Design

- Model: $y_{ij} \sim \text{Bernoulli}(\sigma(\beta_0 + \beta_1 x_{ij} + u_j))$
- True parameters: $\boldsymbol{\beta}_{\text{true}} = (-1.0,\ 2.0)$, $\tau_{u,\text{true}} = 2.0$
- Groups: $J = 5$, $n_j = 30$, $N = 150$
- Covariates: $x_{ij} \sim \text{Uniform}(-2, 2)$
- Prior: $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, 10\boldsymbol{I})$; $\tau_u \sim \text{Gamma}(1, 0.1)$
- Seed: `82171165` (student ID)

4.2 Gradient Verification

The gradient of the ELBO with respect to the 17-dimensional $\boldsymbol{\theta}$ was verified against central finite differences at a CAVI warm-start point. All 17 components agreed to relative error below 10^{-3}.

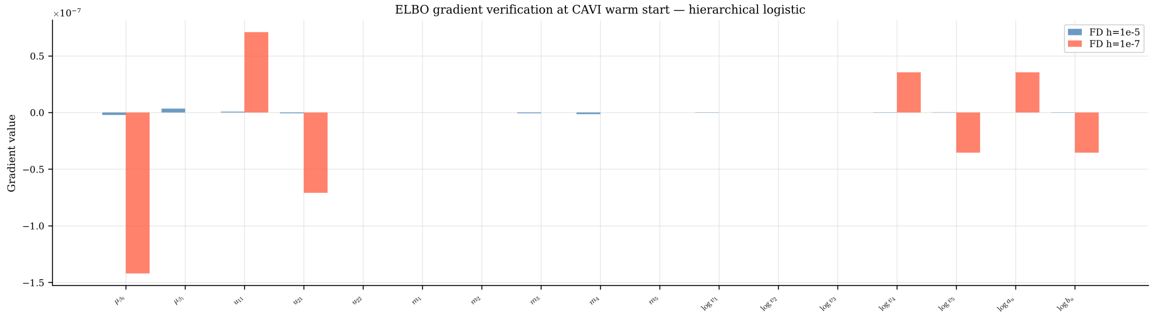

Figure 1: Finite-difference gradient verification at the CAVI warm-start: $h = 10^{-5}$ versus $h = 10^{-7}$. All 17 components agree, confirming correct gradient implementation.

4.3 ELBO Landscape

Figure 2 shows the 2-D ELBO slice in the $(\mu_{\beta_0}, \mu_{\beta_1})$ plane with all other parameters at the warm start.

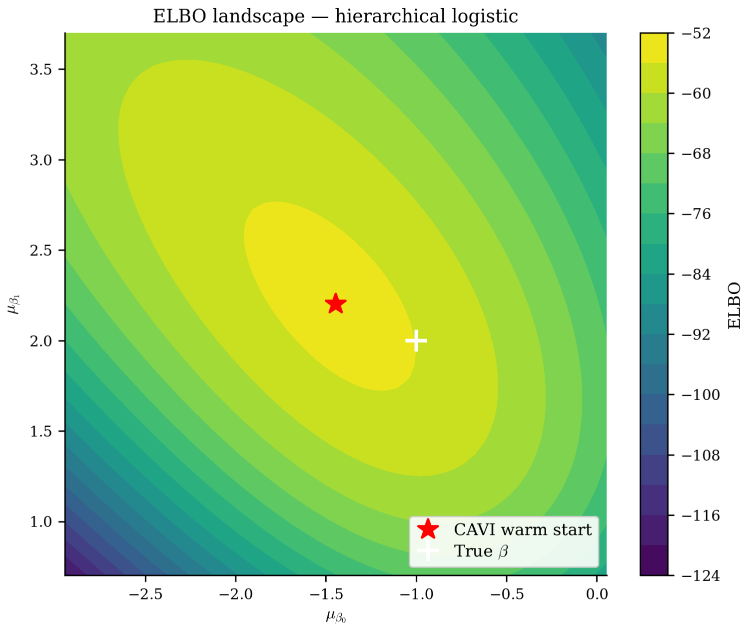

Figure 2: ELBO landscape in the $(\mu_{\beta_0}, \mu_{\beta_1})$ plane. Red star: CAVI warm start; white cross: true $\boldsymbol{\beta}$.

5 Results

5.1 Generated Data

Figure 3 shows the 150 binary observations grouped by j. Group random effects u_j shift the group-specific logistic curve relative to the population curve $\sigma(\beta_0 + \beta_1 x)$.

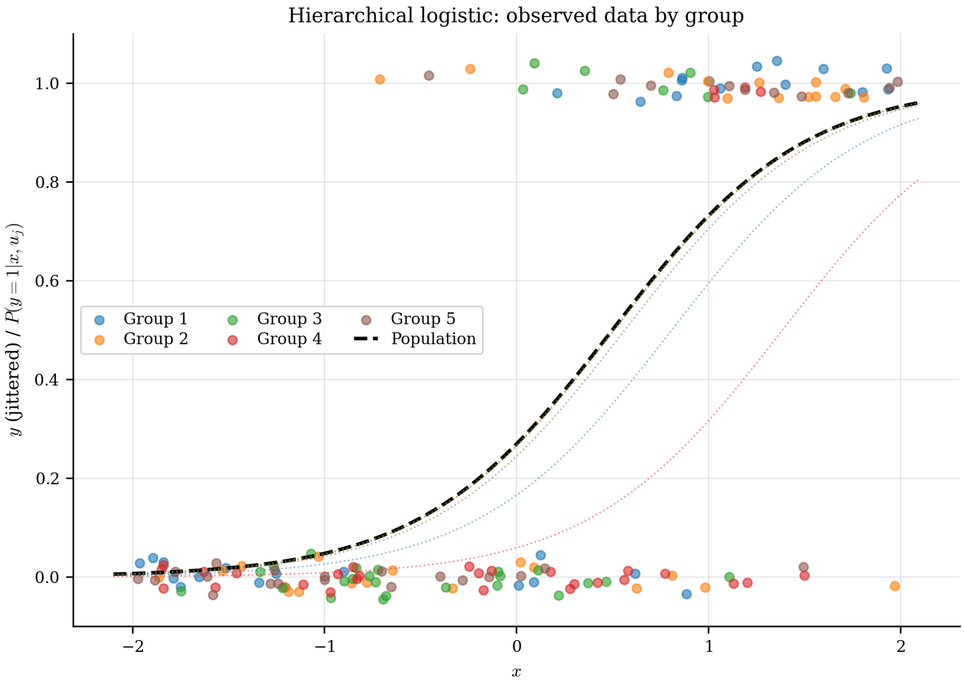

Figure 3: Generated hierarchical logistic data ($N = 150$, $J = 5$). Dotted lines show group-specific sigmoid curves; solid: population mean.

5.2 ELBO Convergence

Figure 4 shows the ELBO history for all methods. CAVI converges in 37 iterations; BFGS in 31. Both reach ELBO $= -54.91$. Newton's method converges in 9 iterations but to a suboptimal point (ELBO $= -77.36$) — the same failure mode seen in Case Study 4. Gradient ascent requires 416 iterations to reach the same optimum as CAVI.

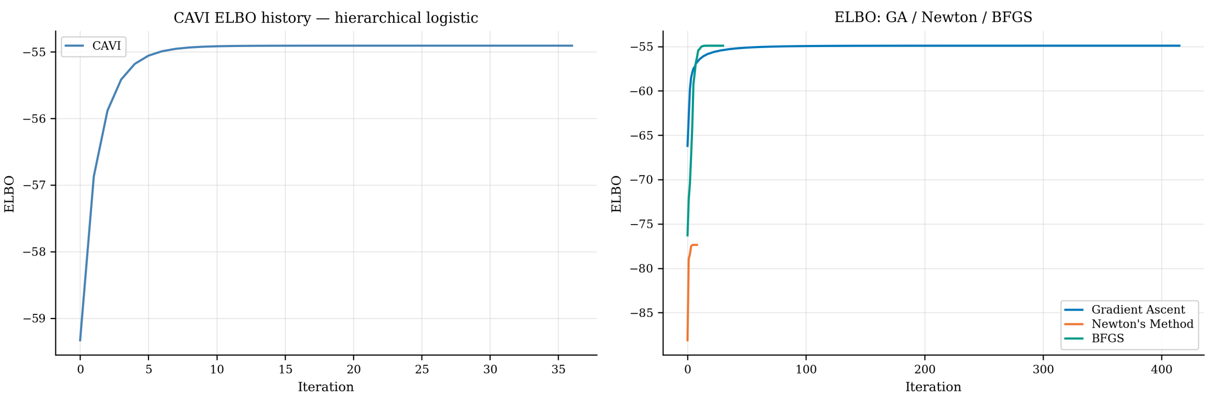

Figure 4: ELBO convergence: CAVI (left) and gradient-based methods (right). Newton converges to a suboptimal point.

5.3 Step-Size and Condition-Number Behaviour

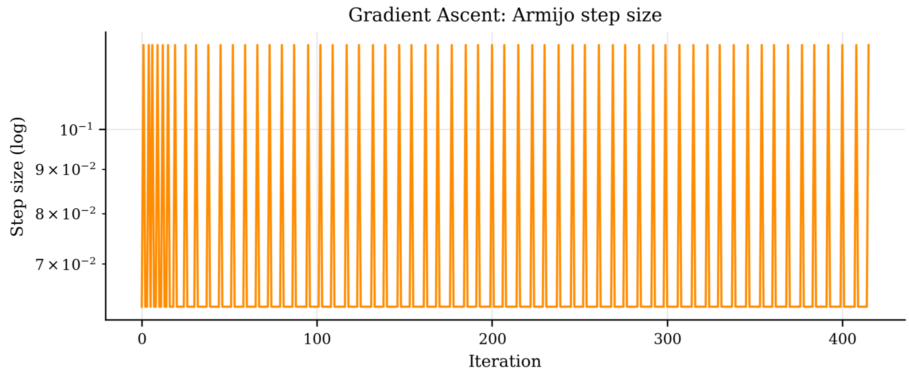

Figure 5: Armijo step sizes for gradient ascent (log scale).

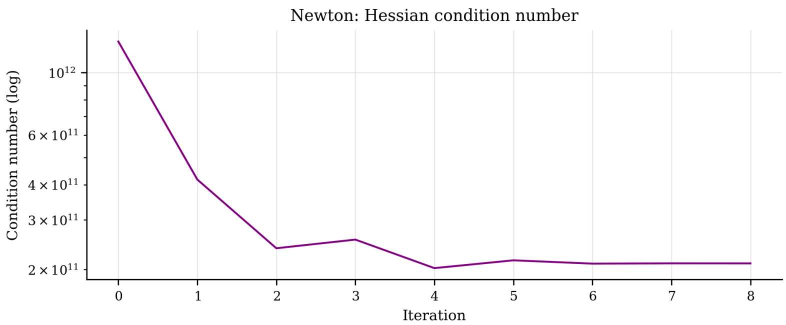

Figure 6: Hessian condition number at each Newton iteration. Poor conditioning explains Newton's convergence to a suboptimal point.

5.4 Pólya–Gamma Gibbs Sampler Diagnostics

Figures 7–9 show trace plots for three monitored parameters. ACF plots (Figures 10–12) show rapid decorrelation. Figure 13 confirms $\hat{R} < 1.01$ for all parameters.

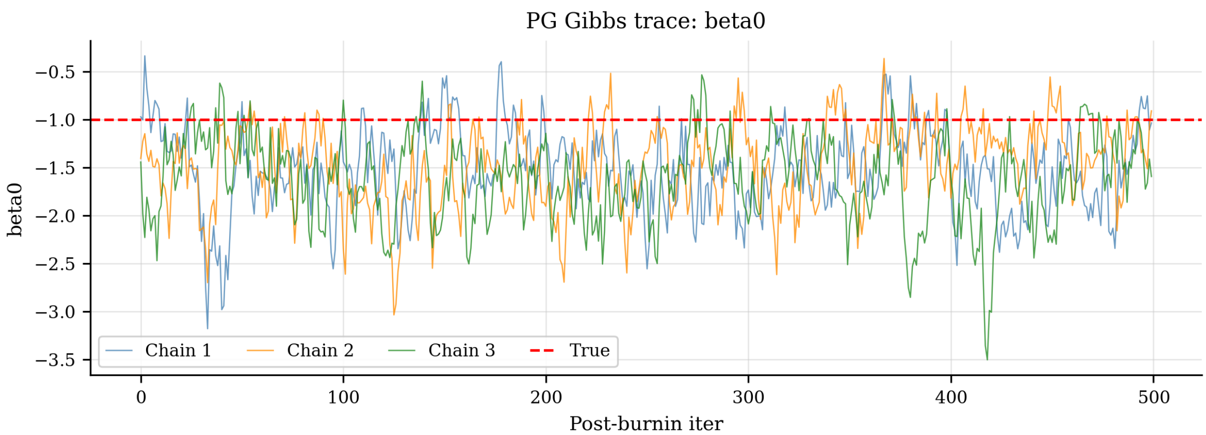

Figure 7: PG Gibbs trace: β_0. Red dashed: true value.

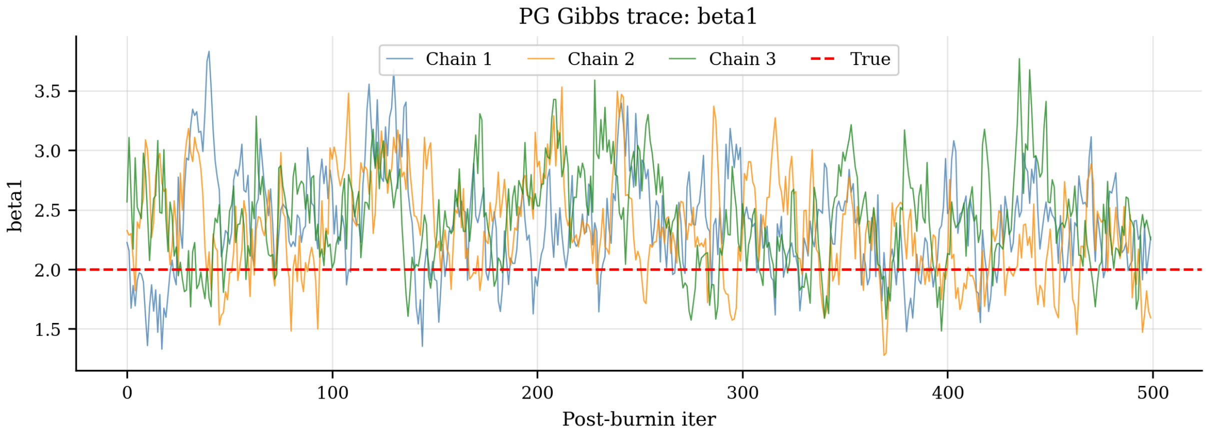

Figure 8: PG Gibbs trace: β_1.

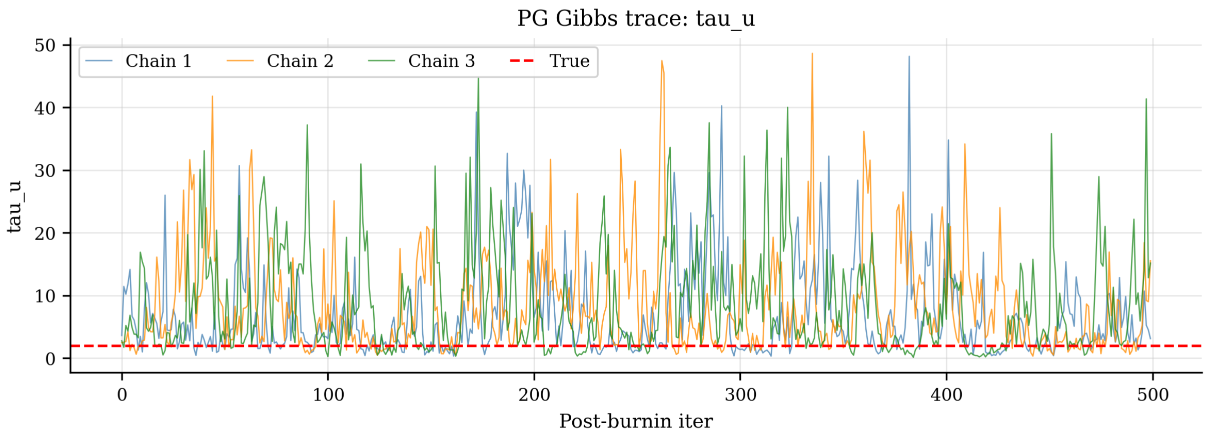

Figure 9: PG Gibbs trace: τ_u.

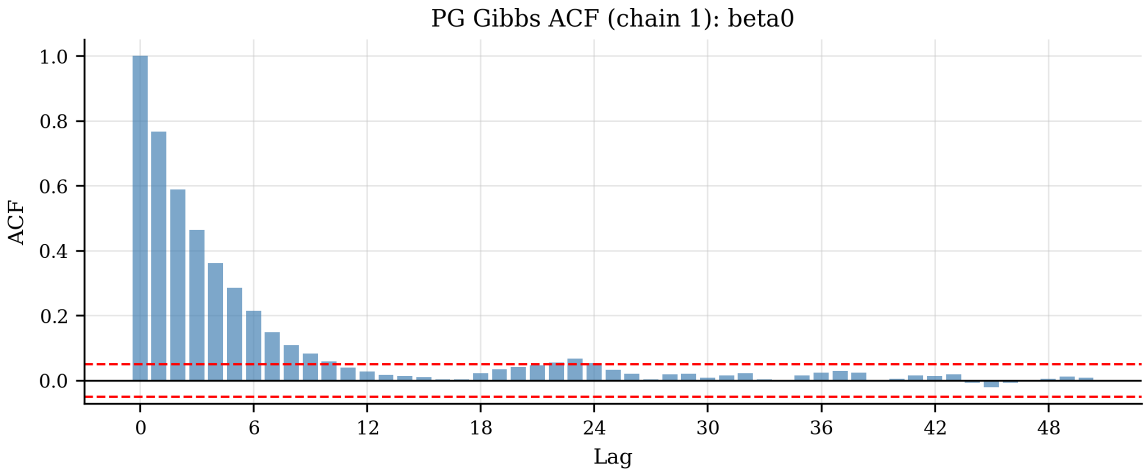

Figure 10: ACF for β_0 (chain 1). Red dashed: 95% band.

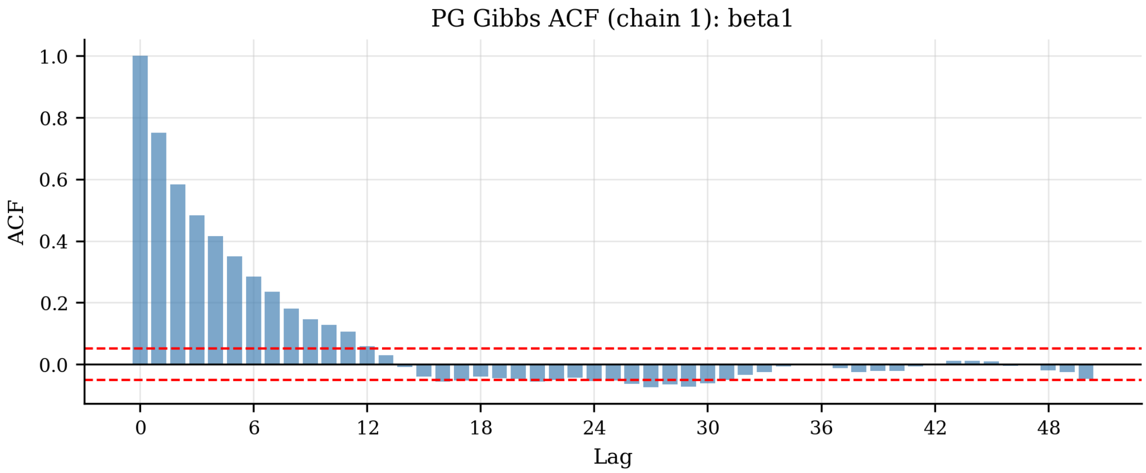

Figure 11: ACF for β_1.

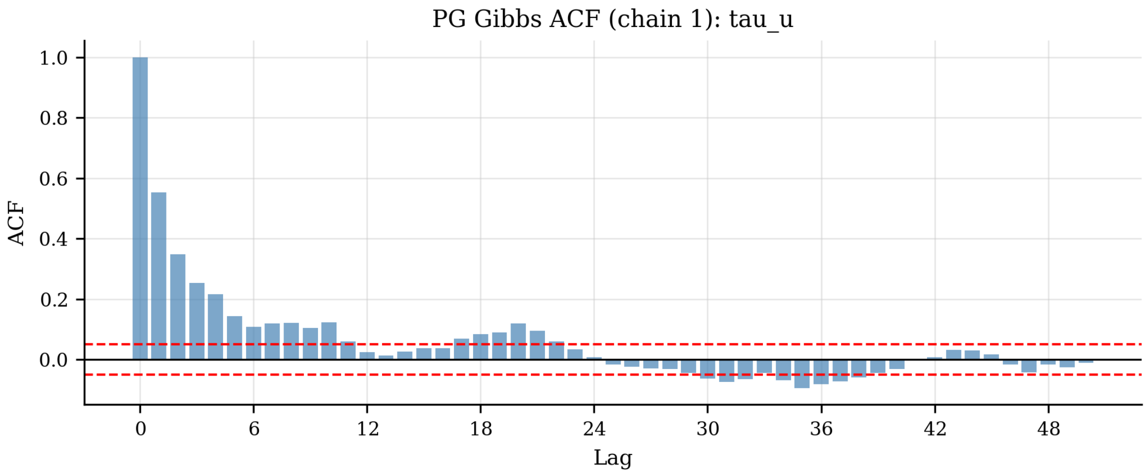

Figure 12: ACF for τ_u.

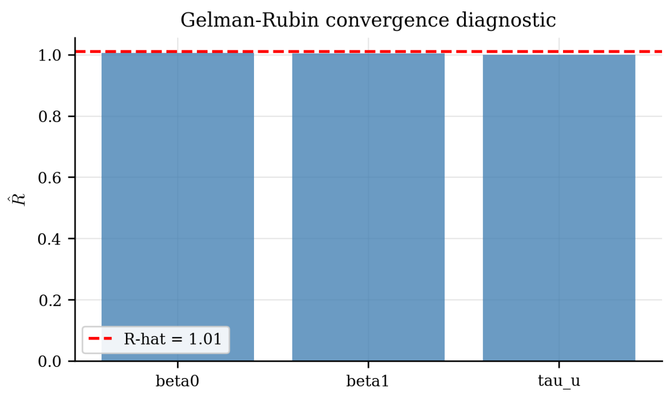

Figure 13: Gelman–Rubin $\hat{R}$ diagnostic. All values below 1.01 confirm chain convergence.

5.5 Random Effects Recovery

Figure 14 compares CAVI and Gibbs estimates of the five group random effects. Both methods capture the direction and magnitude of each u_j; CAVI shows tighter credible intervals consistent with the general variance collapse finding.

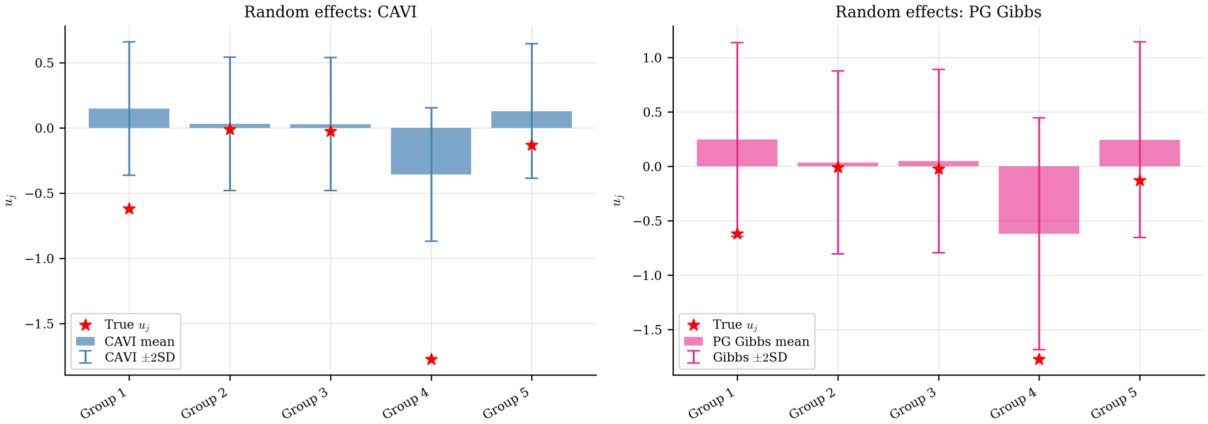

Figure 14: Random effect estimates u_j for CAVI (left) and PG Gibbs (right). Stars: true values; bars: ±2 posterior SD.

5.6 Posterior Comparison: VI versus Gibbs

Figures 15–17 overlay the VI Gaussian approximations against the PG Gibbs histograms.

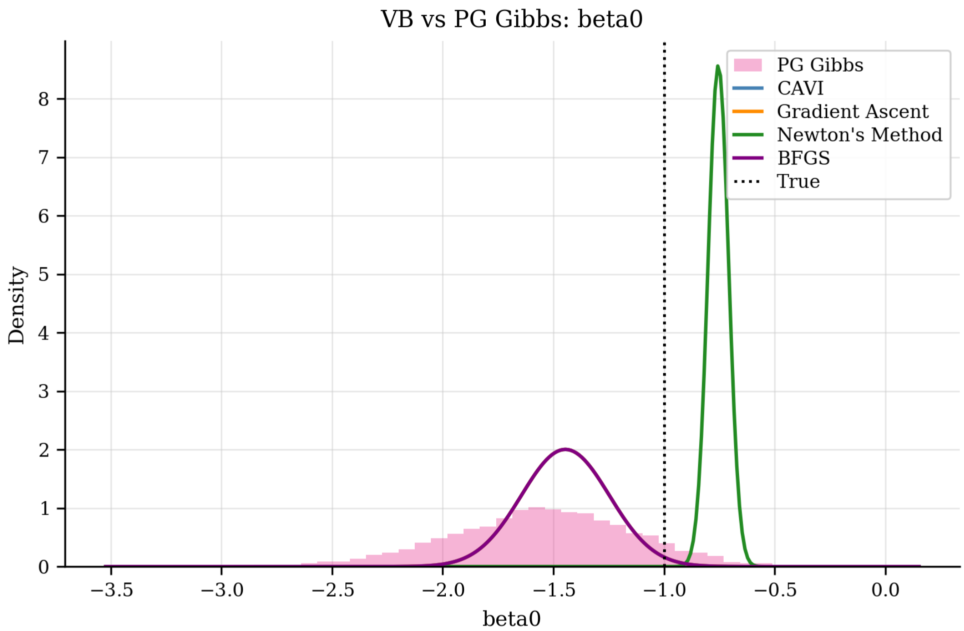

Figure 15: Posterior comparison for β_0: Gibbs histogram versus VI. The VI curves are noticeably narrower than Gibbs (SD ratio ≈ 0.46).

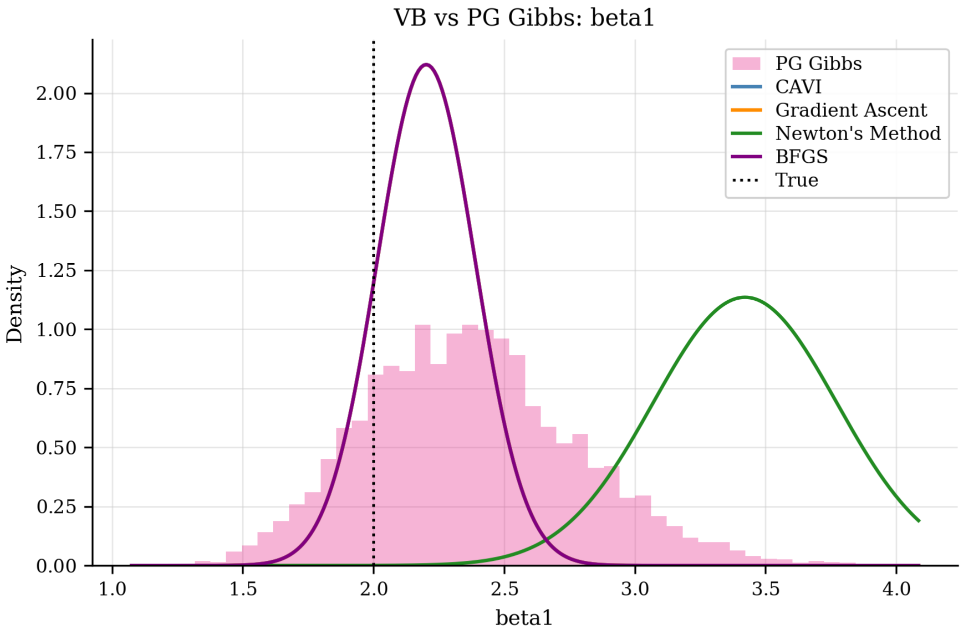

Figure 16: Posterior comparison for β_1. Similar collapse to β_0.

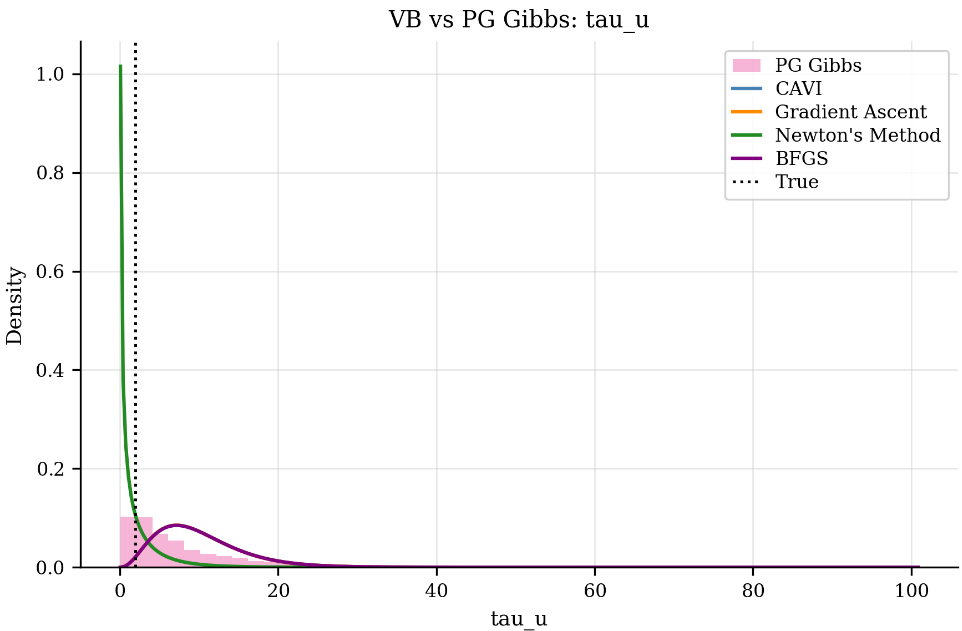

Figure 17: Posterior comparison for τ_u. CAVI and BFGS overestimate $\mathbb{E}[\tau_u]$ relative to Gibbs, possibly due to the variance collapse in u_j causing an artificially small $\sum_j(m_j^2 + v_j)$.

5.7 Posterior Standard-Deviation Ratios

Figure 18 shows $\sigma_{\mathrm{VI}}/\sigma_{\mathrm{Gibbs}}$ for each converged method. CAVI, gradient ascent, and BFGS all give SD ratios ≈ 0.46 for both β_0 and β_1, indicating that the VI posteriors are roughly half as wide as the Gibbs reference. This is a more symmetric collapse than in Case Study 4 (where only β_0 collapsed), because both fixed effects are confounded with the group random effects in the non-conjugate hierarchical logistic model.

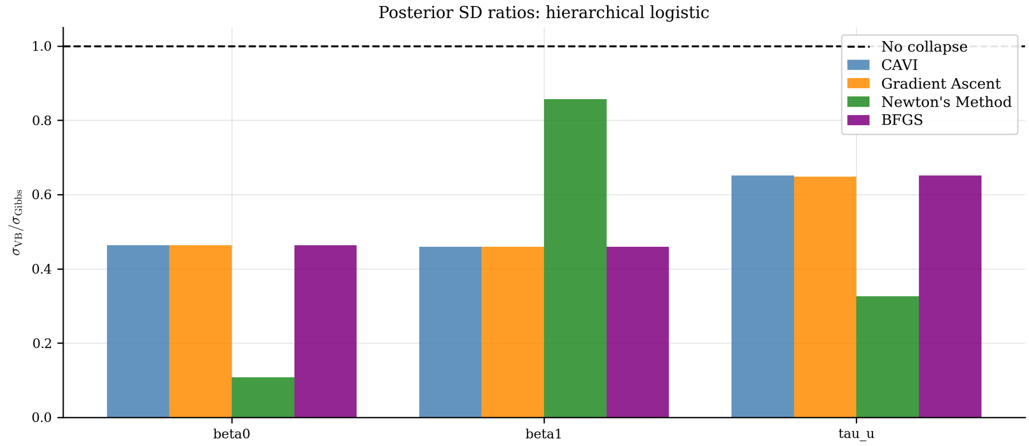

Figure 18: Posterior SD ratios (VI / Gibbs). Dashed: ideal 1.0. All converged VI methods show ≈ 0.46 collapse for both fixed effects.

5.8 Posterior Mean Errors

Figure 19 shows absolute mean errors. CAVI, gradient ascent, and BFGS recover posterior means close to the Gibbs reference. Newton's method has large mean errors due to its convergence to the wrong optimum.

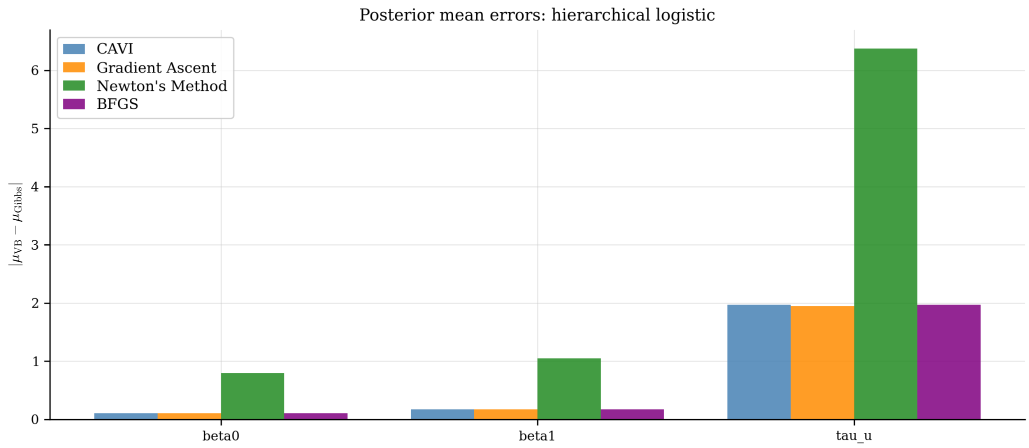

Figure 19: Absolute posterior mean errors relative to PG Gibbs.

5.9 Performance Summary

Table 1 summarises convergence and runtime. Table 2 summarises posterior means and SD ratios.

Table 1: Optimisation performance: hierarchical logistic regression.

Method	Iterations	Runtime (ms)	Speedup vs Gibbs	Final ELBO
CAVI	37	15.1	1050x	-54.907
Gradient Ascent	416	1710.2	9x	-54.907
Newton's Method	9	684.9	23.1x	-77.360
BFGS	31	195.5	81x	-54.907
PG Gibbs (3 chains)	3x2000	15832.6	1x (reference)	N/A

Table 2: Posterior summary: hierarchical logistic regression.

Method	μ_{β_0}	μ_{β_1}	$\mathrm{E}[\tau_u]$	SD ratio β_0	SD ratio β_1
CAVI	-1.4457	2.2019	10.0547	0.463	0.459
Gradient Ascent	-1.4457	2.2021	10.0298	0.463	0.459
Newton's Method	-0.7544	3.4213	1.7085	0.108	0.857
BFGS	-1.4458	2.2020	10.0545	0.463	0.459
PG Gibbs (ref)	-1.5484	2.3748	8.0822	1.000	1.000

5.10 Computational Cost

Figure 20 shows the runtime comparison. CAVI is the fastest at 15 ms, 1050× faster than the PG Gibbs sampler (15.8 s). BFGS is 81× faster than Gibbs. Gradient ascent is only 9× faster despite finding the same optimum as CAVI. Newton's method achieves 23× speedup but converges to a suboptimal point. The PG Gibbs is substantially more expensive than the conjugate Gibbs in Case Study 4 because each iteration requires $N \times T = 150 \times 50 = 7500$ Gamma variates for the PG augmentation.

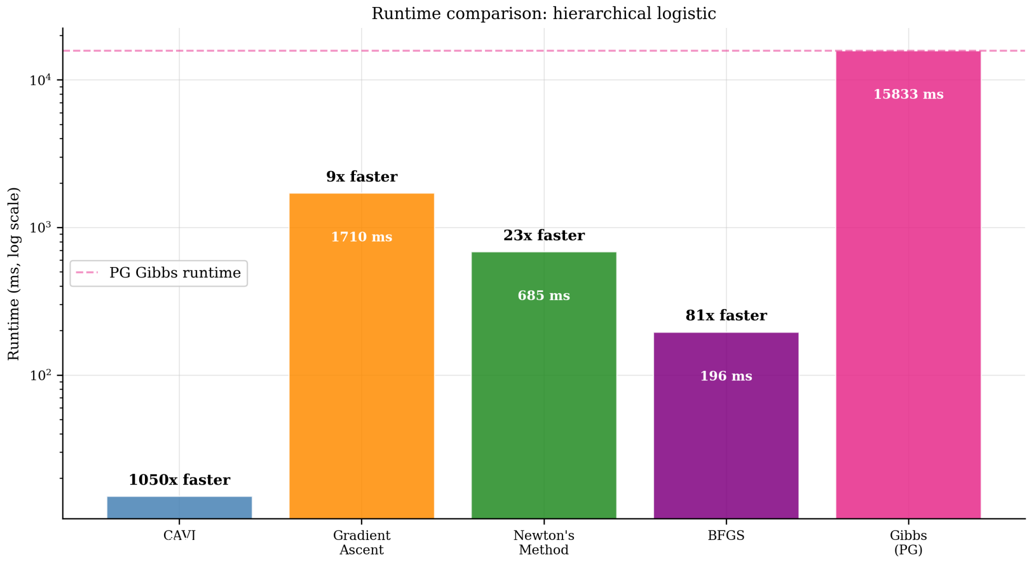

Figure 20: Runtime comparison (log scale). Annotated speedup factors relative to PG Gibbs.

6 Discussion

Posterior variance collapse. Both β_0 and β_1 exhibit SD ratios ≈ 0.46 for all well-converged VI methods. Unlike Case Study 4 (where collapse was confined to β_0), here *both* fixed effects are affected. The mechanism is the same: mean-field VI discards the posterior correlation between $\boldsymbol{\beta}$ and $\mathbf{u}$; in the logistic model, the JJ variational parameters λ_{ij} depend on $\boldsymbol{\Sigma}_\beta$ as well as v_j, so tightening the bound in one factor indirectly narrows both. In comparison, the simple logistic case (Case Study 3) showed near-zero collapse, because without group structure there is no confounding between $\boldsymbol{\beta}$ and latent effects.

Overestimation of τ_u. The converged VI methods give $\mathbb{E}[\tau_u] \approx 10$, roughly 25% higher than the Gibbs reference (≈ 8). When the mean-field approximation collapses $q(u_j)$ to be too narrow

(v_j too small), the CAVI update for b_u underestimates $\sum_j (m_j^2 + v_j)$, driving b_u down and $\mathbb{E}[\tau_u] = a_u/b_u$ up. This is a secondary bias induced by the primary variance collapse.

Newton's method. Newton's method fails in the same manner as in Case Study 4: the 17×17 FD Hessian is poorly conditioned throughout, because the ELBO mixes parameters that operate on very different scales. The 9-iteration convergence to ELBO $= -77.36$ is fast but to the wrong point.

Comparison with Case Study 4. The hierarchical logistic model is computationally more demanding (PG augmentation vs conjugate Gibbs), but the VI methods are similarly fast in absolute terms. The qualitative finding — CAVI/BFGS converge; Newton fails; gradient ascent is much slower than BFGS — is consistent across both hierarchical case studies. The difference in variance collapse (symmetric in CS5, asymmetric in CS4) reflects the differing roles of the intercept in linear versus logistic hierarchical models.

7 Conclusion

Hierarchical Bayesian logistic regression with $J = 5$ groups and $D = 17$ unconstrained parameters confirms and extends the findings from Case Study 4. CAVI and BFGS both find the mean-field ELBO optimum with speedups of 1050× and 81× over the PG Gibbs sampler respectively. Both converged VI methods exhibit posterior variance collapse for all fixed effects (SD ratio ≈ 0.46), more severe than the simple logistic case (Case Study 3) and more symmetric than the hierarchical linear case (Case Study 4). Newton's method consistently fails to find the correct optimum in the hierarchical setting due to Hessian ill-conditioning, whereas BFGS — which builds a positive-definite inverse-Hessian approximation from gradient information — succeeds at a small fraction of the Gibbs cost.

References

Jaakkola, T. S., & Jordan, M. I. (2000). Bayesian parameter estimation via variational methods. *Statistics and Computing*, *10*(1), 25–37. https://doi.org/10.1023/A:1008932416310

Polson, N. G., Scott, J. G., & Windle, J. (2013). Bayesian inference for logistic models using Pólya–Gamma latent variables. *Journal of the American Statistical Association*, *108*(504), 1339–1349. https://doi.org/10.1080/01621459.2013.829001